



Darshan
UNIVERSITY

Python Programming - 2301CS404

Lab - 15 (1)

Roll No.: 459

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01) Write a Program to create a class by name Students, and initialize attributes like name, age, and grade while creating an object.

```
In [1]: class Students :
        def __init__(self,name,age,grade) :
            self.name = name
            self.age = age
            self.grade = grade

        def __str__(self):
            return f" name : {self.name}\n age : {self.age}\n grade : {self.grade}"

s1 = Students("vi",19,"A++")
print(s1)
```

```
name : vi
age : 19
grade : A++
```

02) Create a class named Bank_Account with Account_No, User_Name, Email,Account_Type and Account_Balance data members. Also create a method GetAccountDetails() and DisplayAccountDetails(). Create main method to demonstrate the Bank_Account class.

```
In [7]: class Bank_Account :

        # def __init__(self,accNo,name,email,accType,accBal) :
        #     self.accNo = accNo
        #     self.name = name
        #     self.email = email
```

```

#     self.accType = accType
#     self.accBal = accBal

def getAccDetails(self):
    arr = input("Enter accNo,name,email,accType,accBal : ").split(" ")
    self.accNo = arr[0]
    self.name = arr[1]
    self.email = arr[2]
    self.accType = arr[3]
    self.accBal = arr[4]

def displayAccDetails(self):
    print("accNo : " , self.accNo)
    print("Name : " , self.name)
    print("email : " , self.email)
    print("accType : " , self.accType)
    print("accBal : " , self.accBal)

# b1 = Bank_Account("7","vi","vi@gmail.com","saving","9999999999")
def main():
    b1 = Bank_Account()
    b1.getAccDetails()
    b1.displayAccDetails()

main()

```

```

accNo : 7
Name : vi
email : sh
accType : sav
accBal : 999

```

03) WAP to create Circle class with area and perimeter function to find area and perimeter of circle.

```

In [13]: import math

class Circle :
    def __init__(self,r) :
        self.r = r

    def area(self):
        return math.pi * self.r * self.r
    def perimeter(self):
        return 2 * math.pi * self.r

c1 = Circle(7)
c1.area()
c1.perimeter()

```

```

Out[13]: 43.982297150257104

```

04) Create a class for employees that includes attributes such as name, age, salary, and methods to update and

display employee information.

```
In [16]: class Employees :

    def __init__(self,name,age,salary,accBal) :
        self.name = name
        self.age = age
        self.salary = salary
        self.accBal = accBal

    def displayEmp(self):
        print("Name :" , self.name)
        print("Age :" , self.age)
        print("Salary :" , self.salary)
        print("AccBal :" , self.accBal)

    def update(self,name,age,salary,accBal) :
        self.name = name
        self.age = age
        self.salary = salary
        self.accBal = accBal

e1 = Employees("vi",19,99999,7777777)

e1.displayEmp()
e1.update("vibhu",19,99999,9999999)
print("\n")
e1.displayEmp()
```

```
Name : vi
Age : 19
Salary : 99999
AccBal : 7777777
```

```
Name : vibhu
Age : 19
Salary : 99999
AccBal : 9999999
```

05) Create a bank account class with methods to deposit, withdraw, and check balance.

```
In [23]: class Bank:

    def __init__(self):
        self.bal = 5000

    def deposit(self,amount):
        self.bal += amount
        print("Deposit : ",amount)

    def withdraw(self,amount):
        self.bal -= amount
        print("Withdraw : ",amount)

    def displayBal(self):
        print("Current : ",self.bal)
```

```
b1 = Bank()
b1.deposit(100)
b1.withdraw(100)
b1.displayBal()
```

Deposit : 100
Withdraw : 100
Current : 5000

06) Create a class for managing inventory that includes attributes such as item name, price, quantity, and methods to add, remove, and update items.

```
In [25]: class inventory:
        def __init__(self,name,price,quantity):
            self.name = name
            self.price = price
            self.quantity = quantity

        def add(self,quantity):
            self.quantity += quantity

        def remove(self,quantity):
            self.quantity -= quantity

        def update(self,name,price,quantity):
            self.name = name
            self.price = price
            self.quantity = quantity

i1 = inventory("vi",9999,10)
i1.add(10)
print(i1.quantity)
i1.remove(5)
print(i1.quantity)
```

20
15

07) Create a Class with instance attributes of your choice.

```
In [ ]: import math

class Circle :
    def __init__(self,r) :
        self.r = r

    def area(self):
        return math.pi * self.r * self.r
    def perimeter(self):
        return 2 * math.pi * self.r

c1 = Circle(7)
c1.area()
c1.perimeter()
```

08) Create one class student_kit

Within the student_kit class create one class attribute principal name (Mr ABC)

Create one attendance method and take input as number of days.

While creating student take input their name .

Create one certificate for each student by taking input of number of days present in class.

```
In [28]: class StudentKit:
    principal = "jay"
    principal2 = "Mohit"

    def __init__(self,name):
        self.name = name
        self.attendance = 0

    def atten(self,days):
        self.attendance += (days/365) * 100

    def certificate(self):

        if(self.attendance > 75):
            print("You Are Certified.")
        else:
            print("Attandance Is Below")

s1 = StudentKit("vi")
s1.atten(290)
s1.certificate()
```

You Are Certified.

09) Define Time class with hour and minute as data member. Also define addition method to add two time objects.

```
In [37]: class Time:
    def __init__(self,h,m,s):
        self.h = h
        self.m = m
        self.s = s

    def add(self,t2):
        total_h = self.h + t2.h
        total_m = self.m + t2.m
        total_s = self.s + t2.s

        if(total_s >= 60):
            total_m += total_s // 60
            total_s = total_s % 60

        if(total_m >= 60):
            total_h += total_m // 60
```

```
        total_m = total_m % 60

        total_h = total_h % 12

    return Time(total_h,total_m,total_s)

t1 = Time(1,20,50)
t2 = Time(1,50,50)
t3 = t1.add(t2)
print(f"{t3.h} : {t3.m} : {t3.s}")
```

3 : 11 : 40